

# Satellite Communications

## ECE 4644 / 5610 Spring 2026

Week #7

March 2 - 6, 2026

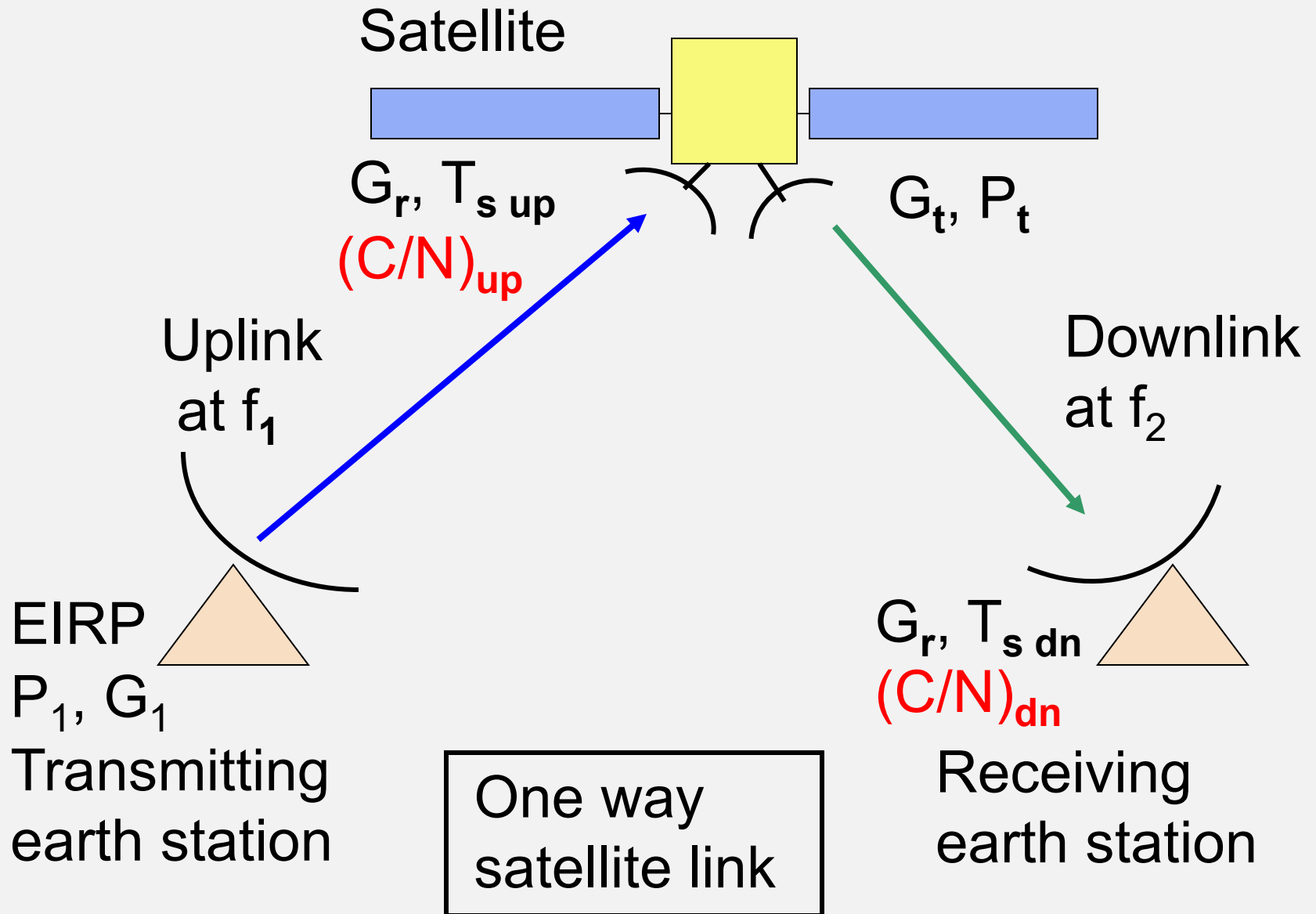
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# Topics

- C/N ratio on satellite links – uplink, downlink and overall C/N
- 1<sup>st</sup> Design exercise – Tx Earth station
- Transponder nonlinearity / Backoff
- Designing satellite communication links
- Example: Intelsat C-band system
- Link margins for rain (Intro)

# C/N on satellite links

- A satellite link consists of an *uplink* from a transmitting earth station to the satellite and a *downlink* from the satellite to a receiving earth station
- Each link has an associated (C/N) ratio, namely  $(C/N)_{\text{up}}$  and  $(C/N)_{\text{dn}}$



# C/N on satellite links

- $(C/N)_{\text{up}}$  and  $(C/N)_{\text{dn}}$  combine into an overall ratio,  $(C/N)_o$ , for the combined link
- $(C/N)_o$  is measured at the output of the IF amplifier or input to the demodulator at the Earth station
- $(C/N)_o$  determines the performance of the combined link
- Intuitively, it is clear that  $(C/N)_o$  cannot be better, i.e., higher, than the lower of  $(C/N)_{\text{up}}$  and  $(C/N)_{\text{dn}}$

# C/N on satellite links: Uplink

- In the transponder we have:

$$(C/N)_{\text{up}} = P_r / k T_{\text{s up}} B_n$$

- $P_r$  is measured at the antenna terminals
- $T_{\text{s up}}$  is the system noise temperature for the transponder
- $B_n$  is the noise bandwidth

# C/N on satellite links: Uplink

- Note:  $B_n$  is the final bandwidth of the *earth station receiver* (not the transponder bandwidth)
- $B_n$  is usually set by the bandwidth of the IF filter that precedes the demodulator
- The earth station IF filter is the ‘chokepoint’ for the link
- We can ignore noise power that falls outside the passband of this final filter before the demodulator

## C/N on satellite links: Downlink

- At the Earth Station we have:

$$(C/N)_{\text{down}} = P_r / k T_{s \text{ dn}} B_n$$

- $P_r$  is measured at the antenna terminals
- $T_{s \text{ dn}}$  is the system noise temperature for the earth station
- $B_n$  is the noise bandwidth of the earth station receiver (IF)

## C/N on satellite links

- Overall C/N ratio at the output of the earth station receiver is given by (Eqn 4.50 – text)

$$1/(C/N)_o = 1/(C/N)_{up} + 1/(C/N)_{dn}$$

- This is called the *reciprocal formula*
- Similar to the equation for the resistance of two resistors in parallel
- C/N values are linear ratios – **NOT in dB**

## Adding C/N Ratios using dB

- If  $(C/N)_{\text{up}} = (C/N)_{\text{dn}}$   
then  $(C/N)_o = (C/N) - 3 \text{ dB}$  (C/N up or dn)
- If  $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 10 \text{ dB}$   
then  $(C/N)_o \sim (C/N)_{\text{dn}} - 0.5 \text{ dB}$
- If  $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 20 \text{ dB}$   
then  $(C/N)_o \sim (C/N)_{\text{dn}} - 0.1 \text{ dB}$
- If  $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 30 \text{ dB}$   
then  $(C/N)_o \sim (C/N)_{\text{dn}}$

## C/N on satellite links

- Example #6

Uplink C/N = 20 dB      ratio 100

Downlink C/N = 20 dB      ratio 100

$$\begin{aligned} 1/(C/N)_o &= 1/(C/N)_{up} + 1/(C/N)_{dn} \\ &= 1 / 100 + 1 / 100 \\ &= 0.02 \end{aligned}$$

$$(C/N)_o = 50 \text{ or } 17.0 \text{ dB}$$

# C/N on satellite links

- Example # 7

Uplink C/N = 30 dB      ratio 1000

Downlink C/N = 20 dB      ratio 100

$$\begin{aligned} 1/(C/N)_o &= 1/(C/N)_{up} + 1/(C/N)_{dn} \\ &= 1 / 1000 + 1 / 100 \\ &= 0.011 \end{aligned}$$

$$(C/N)_o = 90.9 \text{ or } 19.6 \text{ dB}$$

# Summary

- $C (= P_r)$ , received signal power) is calculated using the link equation
- $N (= P_n)$ , noise power in receiver) is calculated using a noise model and the  $kT_s B_n$  relationship
- Rain in a satellite link path causes attenuation and, on the downlink, increases noise temperature
- $(C/N)_o$  determines the performance of the overall link via the reciprocal formula



# How to Design a Satellite Communication System (Section 4.8 of text)

For two-way links, there are four separate links to be designed:

Outbound: uplink and downlink

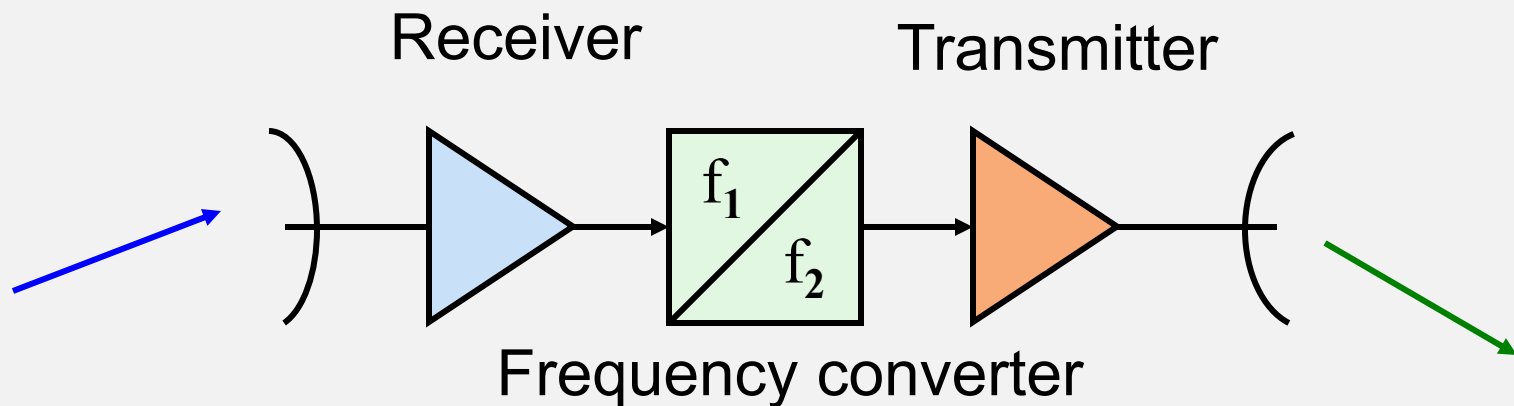
Inbound: uplink and downlink

Two cases: clear air, rain



# Transponders

- Basic transponder (bent pipe)



Uplink at Frequency  $f_1$

Downlink at frequency  $f_2$

# Transponders

- Two Types:
- Bent pipe (linear)
  - Signal is received, amplified
  - Frequency changed, amplified, retransmitted
- On-Board Processing (OBP)
  - Signal is received, amplified, demodulated
  - Bit stream is processed
  - Bits are re-modulated, transmitted

# Uplinks

- Choose your band: L, C, Ku, Ka
- L band not affected by rain
- C band affected only by very heavy rain
- Ku band will have outages due to rain
- Ka band will suffer more outages than Ku band – attenuation is four times greater than Ku band for same rain rate

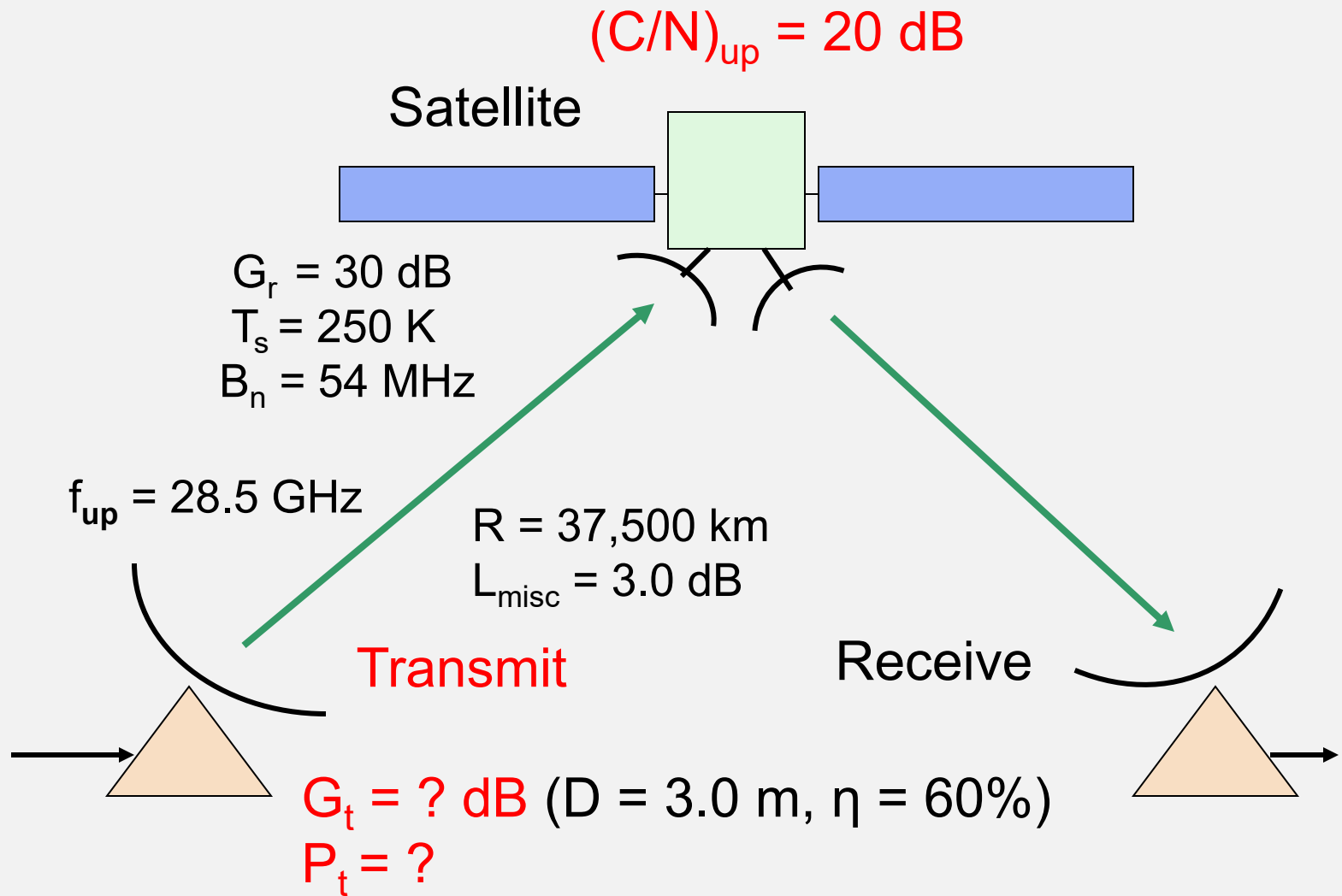
# 1<sup>st</sup> Design Exercise – Tx Earth Station

- Design a Tx Earth station to achieve **C/N = 20 dB** in a Ka band transponder ( $f_{\text{up}} = 28.5$  GHz)
- Orbit is GEO, take distance to be 37,500 km
- Uplink antenna is circular with a diameter of 3 m and an aperture efficiency of 60%
- The receive antenna has a gain of 30 dB on its axis, the receive system noise temperature is 250 K, and the noise bandwidth is 54 MHz

# 1<sup>st</sup> Design Exercise – Tx Earth Station

- Allow for 3.0 dB for clear air attenuation and miscellaneous losses
- Determine the transmitter power,  $P_t$ , that is needed to achieve the desired C/N in the transponder
- We will apply the link equation in dB form:

$$P_r = P_t + G_t + G_r - L_p - L_{tot} \quad \text{dBW}$$



## Design of Ka-band Tx Earth Station

## Tx Earth Station

- Find the wavelength from

$$f \text{ [GHz]} \lambda \text{ [cm]} = 30$$

$$\text{with } f = 28.5 \text{ GHz}$$

$$\lambda = 1.0526 \text{ cm} = 0.010526 \text{ m}$$

- Find the gain,  $G_t$ , of the transmit antenna from the exact expression

$$\begin{aligned} G &= \eta_A \times 4 \pi A / \lambda^2 = (0.60) \times 4 \pi (\pi 1.5^2) / \lambda^2 \\ &= 41024 = 56.8 \text{ dB} \end{aligned}$$

## Tx Earth Station

- Find the path loss,  $L_p$  (in dB), from

$$\begin{aligned} L_p &= 10 \log [(4\pi R / \lambda)^2] \\ &= 10 \log [(4\pi 37,500 \cdot 10^3 / 0.010526)^2] \end{aligned}$$

(note conversion of distances to meters)

$$= 213.0 \text{ dB}$$

- Find the total (non-path) losses,  $L_{\text{tot}}$ , as just

$$L_{\text{tot}} = L_{\text{misc}} = 3.0 \text{ dB}$$

## Tx Earth Station

- We are now ready to enter the link budget equation (modified for a loss term):

$$P_r = P_t + G_t + G_r - L_p - L_{\text{tot}} \quad \text{dBW}$$

- It is convenient to arrange this in tabular form (like a household budget)

# Ka-band Uplink

- Uplink power budget

|                        |                   |
|------------------------|-------------------|
| $P_t$ at earth station | $P_t$ dBW         |
| $G_t$ at earth station | 56.8 dB           |
| $G_r$ at satellite     | 30.0 dB           |
| $L_p$                  | -213.0 dB         |
| $L_{tot}$              | <u>-3.0 dB</u>    |
| $P_r$ at satellite     | $P_t - 129.2$ dBW |

## Tx Earth Station

- Now we have an expression for  $P_r$  at the satellite in terms of the Earth station transmit power,  $P_t$
- We will relate  $P_r$  and hence  $P_t$  to the desired C/N at the satellite
- We need to know the noise power,  $N$ , at the transponder - apply

$$P_n = k T_s B_n$$

# Ka-band Uplink

- Uplink noise power budget

$$k = 1.39 \times 10^{-23} \text{ J/K} \quad - 228.6 \text{ dBW}$$

$$T_s = 250 \text{ K} \quad 24.0 \text{ dBK}$$

$$B_n = 54 \text{ MHz} \quad \underline{77.3 \text{ dBHz}}$$

$$N \quad - 127.3 \text{ dBW}$$

# Ka-band Uplink

- Now we can write an expression for C / N in dB units:

$$C/N = P_r - N = 20 \text{ dB}$$

- Which implies that

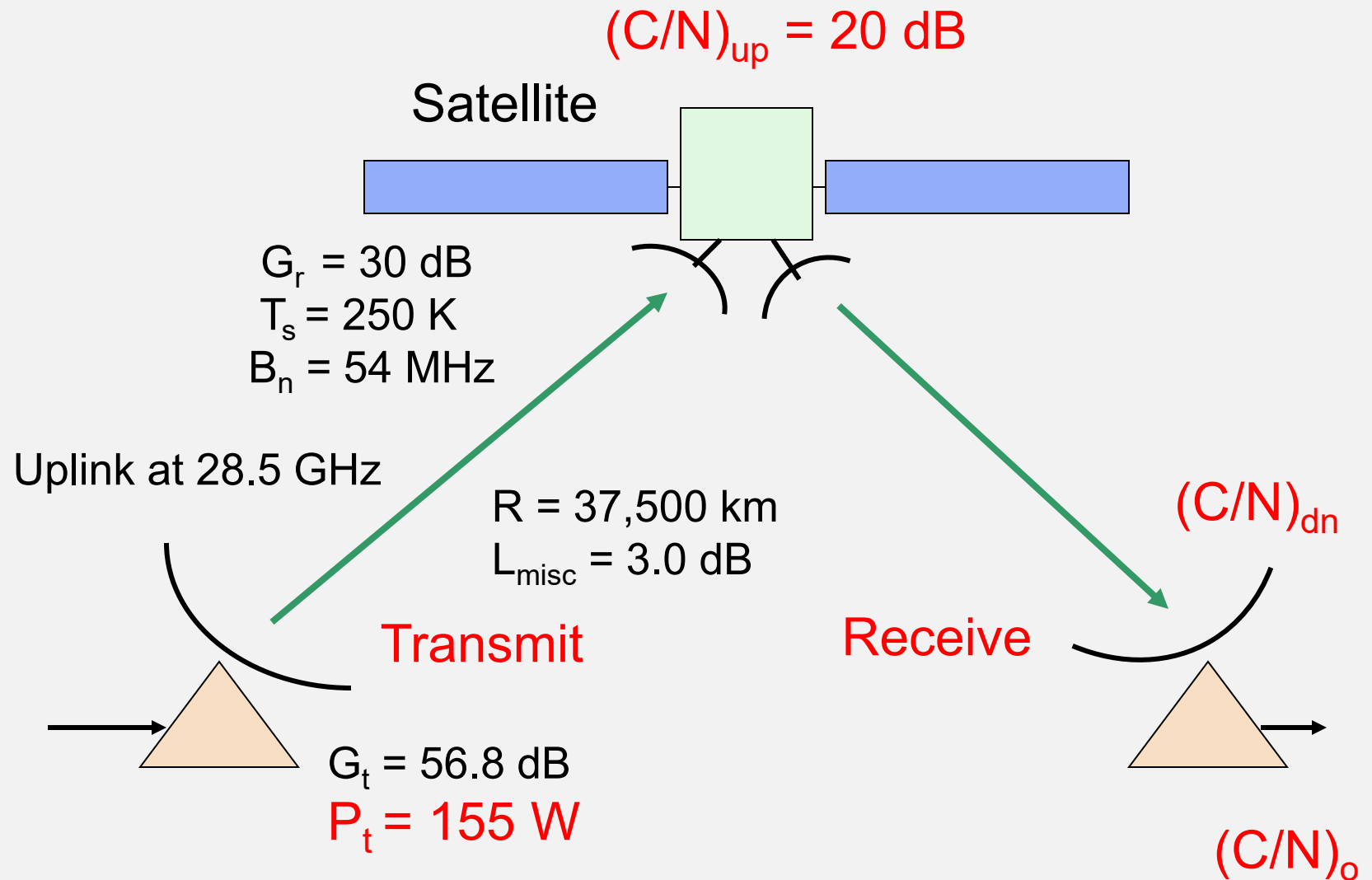
$$\begin{aligned} P_r &= N + 20 \text{ dB} = -127.3 + 20 = -107.3 \text{ dBW} \\ &= P_t - 129.2 \text{ dBW} \text{ (from uplink budget)} \end{aligned}$$

- So

$$P_t = 21.9 \text{ dBW} = 155 \text{ W}$$

# 1<sup>st</sup> Design Exercise

- We have found the Earth Station Tx power that is needed to achieve the desired C/N in the transponder
- Next step: analyze the downlink to find an expression for  $(C/N)_{dn}$
- Then combine  $(C/N)_{up}$  and  $(C/N)_{dn}$  to obtain an expression for  $(C/N)_o$  using the reciprocal formula



**Design of Ka-band link**

$$1/(C/N)_o = 1/(C/N)_{up} + 1/(C/N)_{dn}$$



# Transponder Non-linearity

- Bent pipe transponders have non-linear output stages
- Rated output power is *saturated power*  $P_{\text{sat}}$
- Transponders should not operate at  $P_{\text{sat}}$  when handling more than one carrier
- The carriers interfere with one another through third order *intermodulation products* (IM) – see Section 6.4.4

# Traveling wave tube amplifier I-O characteristic

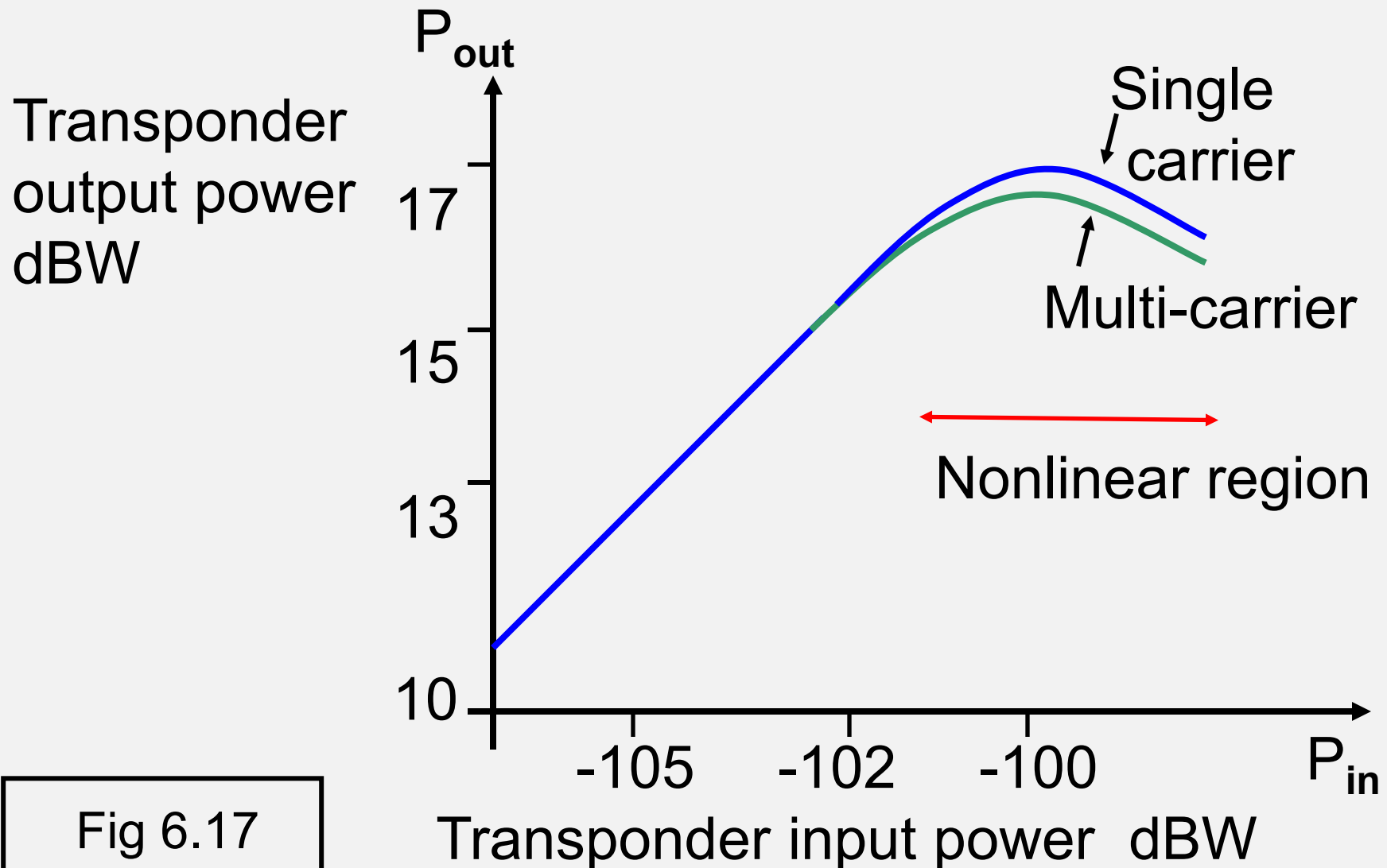


Fig 6.17

# Transponder Non-linearity

- Nonlinearity of the amplifier near saturation increases IM products
- *Back off* must be applied to the output power
- Operating output power =  $P_t$
- Output back off =  $P_{\text{sat}} - P_t$
- Defines an *Operating Point* for the transponder in the  $P_{\text{out}}$  versus  $P_{\text{in}}$  plot

# Traveling wave tube amplifier I-O characteristic

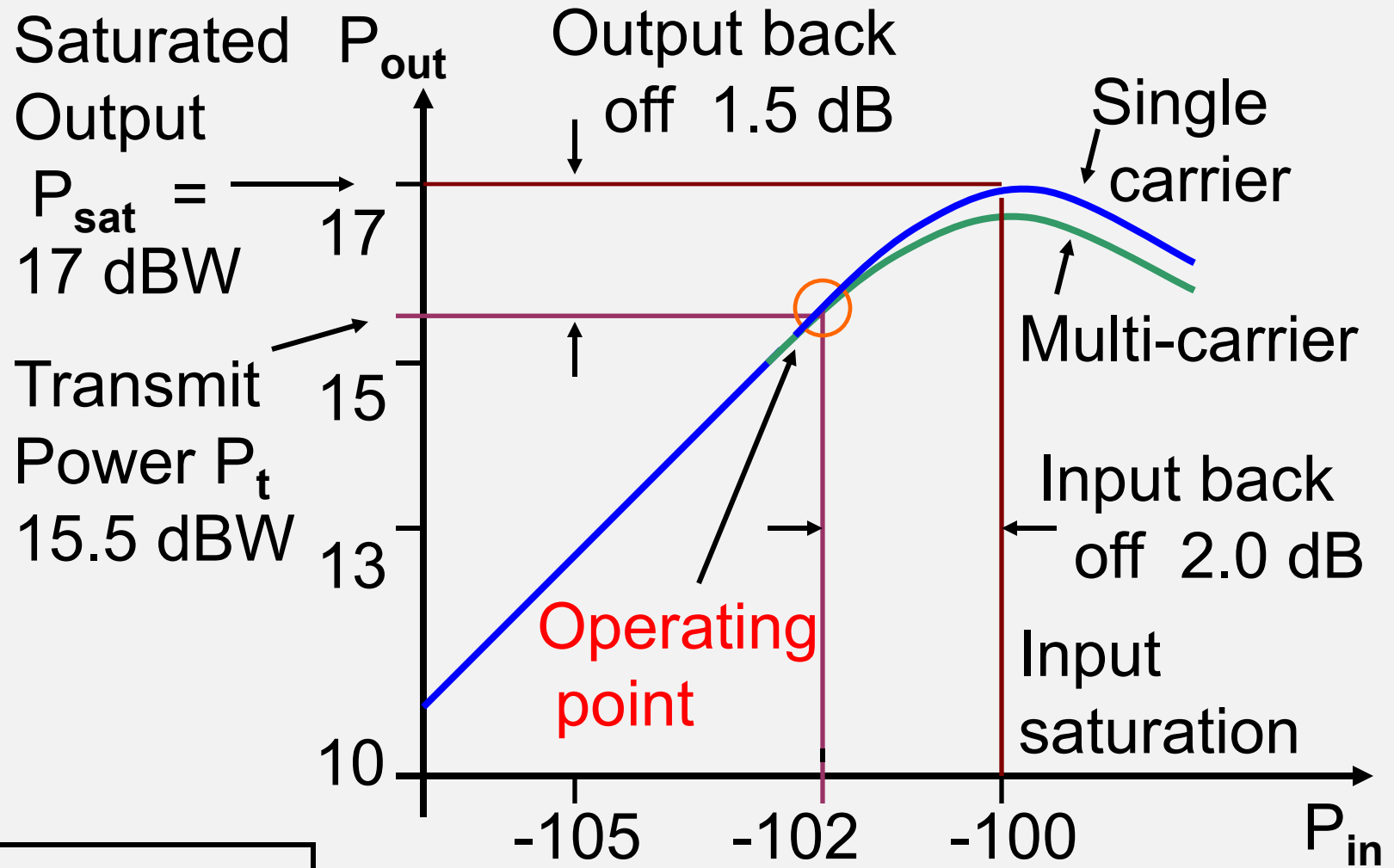


Fig 6.17

Transponder input power dBW

# Back Off

- Input to transponder is also backed off
- Input back off is always larger than output back off
- In Fig 6.17, output back off is 1.5 dB
- Input back off is 2.0 dB
- Transponder gain is  $G_{xp} = P_t - P_{in}$  dB
- Transponder gain in Fig 6.17 is

$$G_{xp} = 15.5 \text{ dBW} - (-102.0 \text{ dBW}) = 117.5 \text{ dB}$$

# Uplinks

- Uplinks:
- Higher frequency than downlink (usually)
- A single uplink earth station may serve many downlink receivers and be highly engineered
- Can achieve high C/N in transponder with large fixed earth station antenna and high power, i.e. high EIRP
- Can use *uplink power control* to combat uplink rain attenuation

# Downlinks

- Often the weaker link in FSS
- Lower frequency than uplink
- Multiple receive terminals requires lower cost per terminal (reduce \$ per customer)  
e.g. VSAT networks, DBS-TV, Starlink
- Small receiving antenna means less signal
- Lower C/N in earth station receiver

# Downlinks

- Example:
- Directv Direct Broadcast Satellite Television
- Two uplink stations
- Millions of receivers
- Receive terminals are major investment
- Must be available to consumers at low price



# Fixed Satellite Service (FSS) Links

- Example: Intelsat link across Atlantic Ocean
- Case of just two earth stations, both highly engineered
- Large C-band antennas at each end of link
  - 30 m diameter
  - $G_t = 63$  dB,  $G_r = 60$  dB
  - $G_r / T_s > 41.7$  dBK<sup>-1</sup> at 5° elevation angle
  - ( $T_s = 18.3$  dBK, 67 K)

# Fixed Satellite Service (FSS) Links

- Beamwidth of antenna is small
- Use  $B/W \sim (33,000 / G)^{1/2}$

At 6 GHz,  $G = 63 \text{ dB} = 2 \times 10^6$

$$\begin{aligned} B/W &= (33,000 / 2 \times 10^6)^{1/2} \\ &= 0.13^\circ \end{aligned}$$

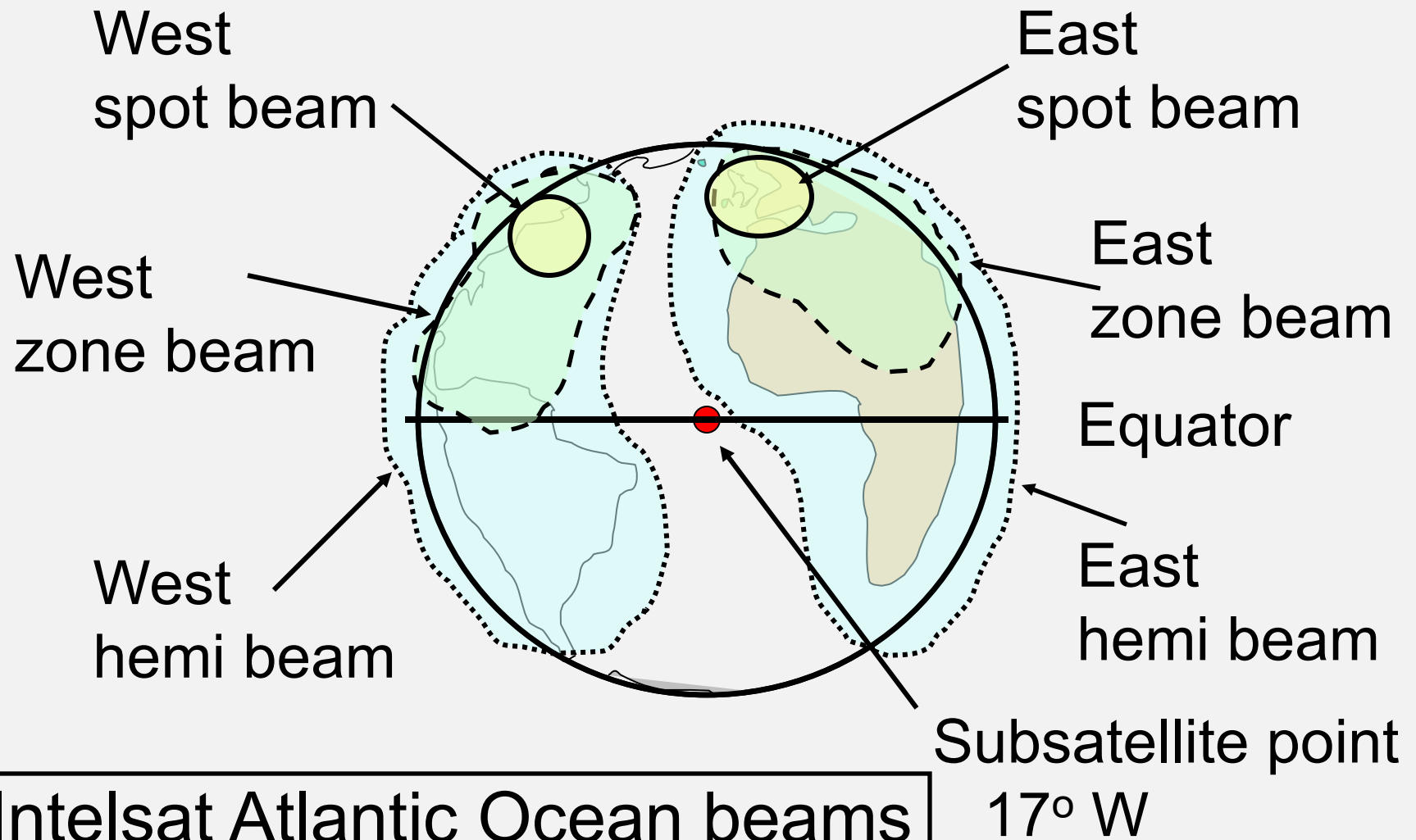
- Narrow beam  $\sim 0.13^\circ$  requires accurate steering to follow motion of satellite ( $\pm 0.05^\circ$ )
- Very expensive antenna (\$10 M)

# Intelsat C-band Uplink

- Intelsat originally designed links with large earth stations
- 30 m antennas and 3 kW transmitters
- High C/N ratios in transponder and receiver
- Very expensive
- C/N in transponder = 30 dB
- Overall C/N = 18.0 dB at receiving earth station
- Guaranteed high quality telephone links

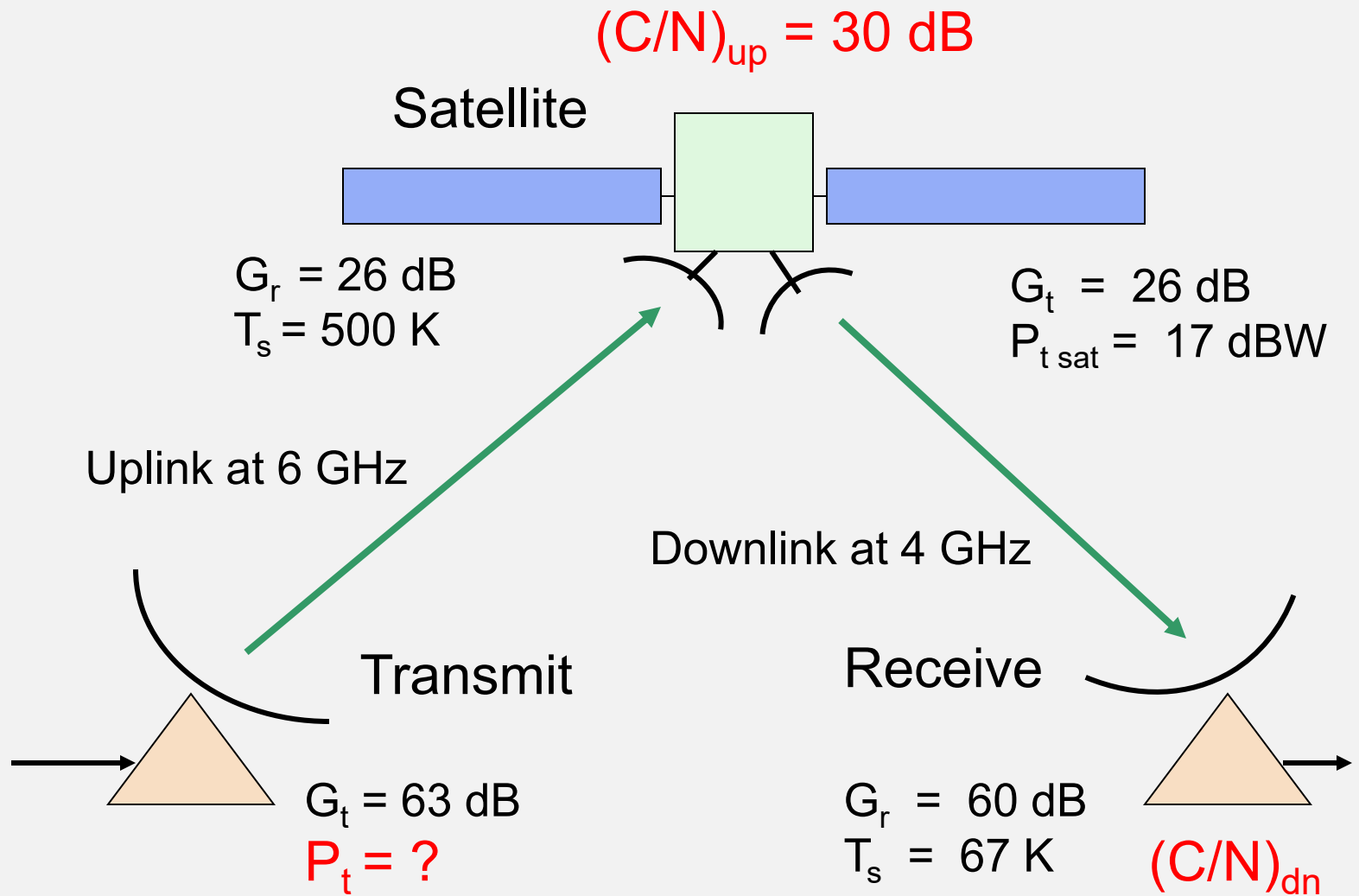
# Intelsat C-band link: Satellite notes

- Satellite transponder power:
  - saturated output power  $50 \text{ W} = 17 \text{ dBW}$
  - apply 2 dB output back-off  
(then  $P_t = 15 \text{ dBW} = 31 \text{ W}$ )
- Receive system noise temperature  $= 500 \text{ K}$
- Satellite zone antenna beam has  $G = 26 \text{ dB}$  on axis
- But design for an earth station on the edge of the beam; apply 3 dB decrease to  $G$



# Intelsat C-band link

- Symmetrical links
- **Required overall  $C/N \geq 18.0$  dB in clear air**
- Design uplink for high  $(C/N)_{up}$  in transponder  
e.g.  $(C/N)_{up} = 30$  dB



Intelsat Atlantic Ocean Region satellite link  $(C/N)_o \geq 18.0 \text{ dB}$

# Intelsat C-band Link Budget (Clear air)

- Uplink:
  - power budget
  - noise power budget
  - C/N ratio
- Downlink:
  - power budget
  - noise power budget
  - C/N ratio
- Overall C/N ratio

# Intelsat C-band Uplink

- Uplink to satellite transponder; solve for required transmit power,  $P_t$
- Find  $P_t$  from  $(C/N)_{up}$
- Set earth station power  $= P_t$  W
- Set  $(C/N)_{up}$  in transponder  $= 30$  dB
- We need expressions for carrier power,  $C$  ( $= P_r$ )
- And for noise power,  $N$ , at the satellite

# Intelsat C-band Uplink

- Uplink power budget

| $P_t$                    | $P_t$ dBW         |
|--------------------------|-------------------|
| $G_t$ at earth station   | 63.0 dB           |
| $L_p$ at 6 GHz (typical) | -200.0 dB         |
| $G_r$ at satellite       | 26.0 dB           |
| Edge of beam loss        | <u>-3.0 dB</u>    |
| $P_r$                    | $P_t - 114.0$ dBW |

# Intelsat C-band Uplink

- Uplink noise power budget

- $k$  -228.6 dBW/K/Hz
- $T_s = 500$  K 27.0 dBK
- $B_n = 36$  MHz 75.6 dBHz
- $N$  -126.0 dBW

- Hence find C/N on uplink as:

$$\begin{aligned}(C/N)_{up} &= P_r - N \\ &= P_t - 114.0 \text{ dBW} - (-126.0 \text{ dBW}) \\ &= P_t + 12.0 \text{ dB}\end{aligned}$$

# Intelsat C-band Uplink

- Uplink C/N ratio
- We require  $(C/N)_{up} = 30$  dB in transponder
- $(C/N)_{up} = P_t + 12.0$  dB = 30.0 dB  
Requires  $P_t = 18$  dBW  
If we include 2 dB of miscellaneous losses  
 $P_t = 20$  dBW = 100 watts
- Note: Earth station EIRP is  
 $EIRP = 20$  dBW + 63 dB = 83 dBW

# Intelsat C-band Downlink

- Satellite transponder
  - Saturated output power  $50\text{ W} = 17\text{ dBW}$
  - Operate with 2 dB output back-off at  $+15\text{ dBW} = 31.6\text{ W}$
  - Back-off is needed for linear operation
  - Set up downlink power budget to receive terminal with Rx antenna gain 60 dB
  - C/N will be high due to high  $G_r$

# Intelsat C-band Downlink

- Downlink power budget

|                          |                 |
|--------------------------|-----------------|
| $P_t$ saturated          | 17.0 dBW        |
| Output back off          | - 2.0 dB        |
| $G_t$ satellite          | 26.0 dB         |
| Downlink path loss (typ) | - 196.0 dB      |
| Receive terminal gain    | 60.0 dB         |
| Edge of beam loss        | - 3.0 dB        |
| Miscellaneous losses     | <u>- 2.0 dB</u> |
| $P_r$                    | - 100.0 dBW     |

# Intelsat C-band Downlink

- Downlink noise power budget

|                  |                  |
|------------------|------------------|
| – k              | - 228.6 dBW/K/Hz |
| – $T_s = 67$ K   | 18.3 dBK         |
| – $B_n = 36$ MHz | <u>75.6 dBHz</u> |
| – N              | - 134.7 dBW      |

- Downlink C/N ratio

$$\begin{aligned} C/N &= -100.0 \text{ dBW} - ( - 134.7 \text{ dBW} ) \\ &= 34.7 \text{ dB} \end{aligned}$$

# Intelsat C-band Link

- Overall C/N ratio is found as the reciprocal sum of 30.0 dB up and 34.7 dB down:

$$1 / (C/N)_o = 0.0010 + 0.000339 = 0.001269$$

$$(C/N)_o = 747 \text{ or } 28.7 \text{ dB}$$

- We have exceeded the minimum required  $(C/N)_o$  value by  $28.7 - 18.0 \approx 10 \text{ dB}$
- Interference, intermodulation, and other factors will reduce  $(C/N)_o$  in practice

# Design Note on Transponder Gain

- Check on gain required of transponder by comparing  $P_r$  and  $P_t$
- Noise power at transponder input was  $N = -126.0$  dBW and  $P_r / N = 30$  dB
- Hence  $P_r = -126.0 + 30.0 = -96.0$  dBW
- Transponder output power  $P_t = +15$  dBW
- Transponder gain required:

$$G_{xp} = +15 - (-96) = 111 \text{ dB} \quad (\text{Ok?})$$



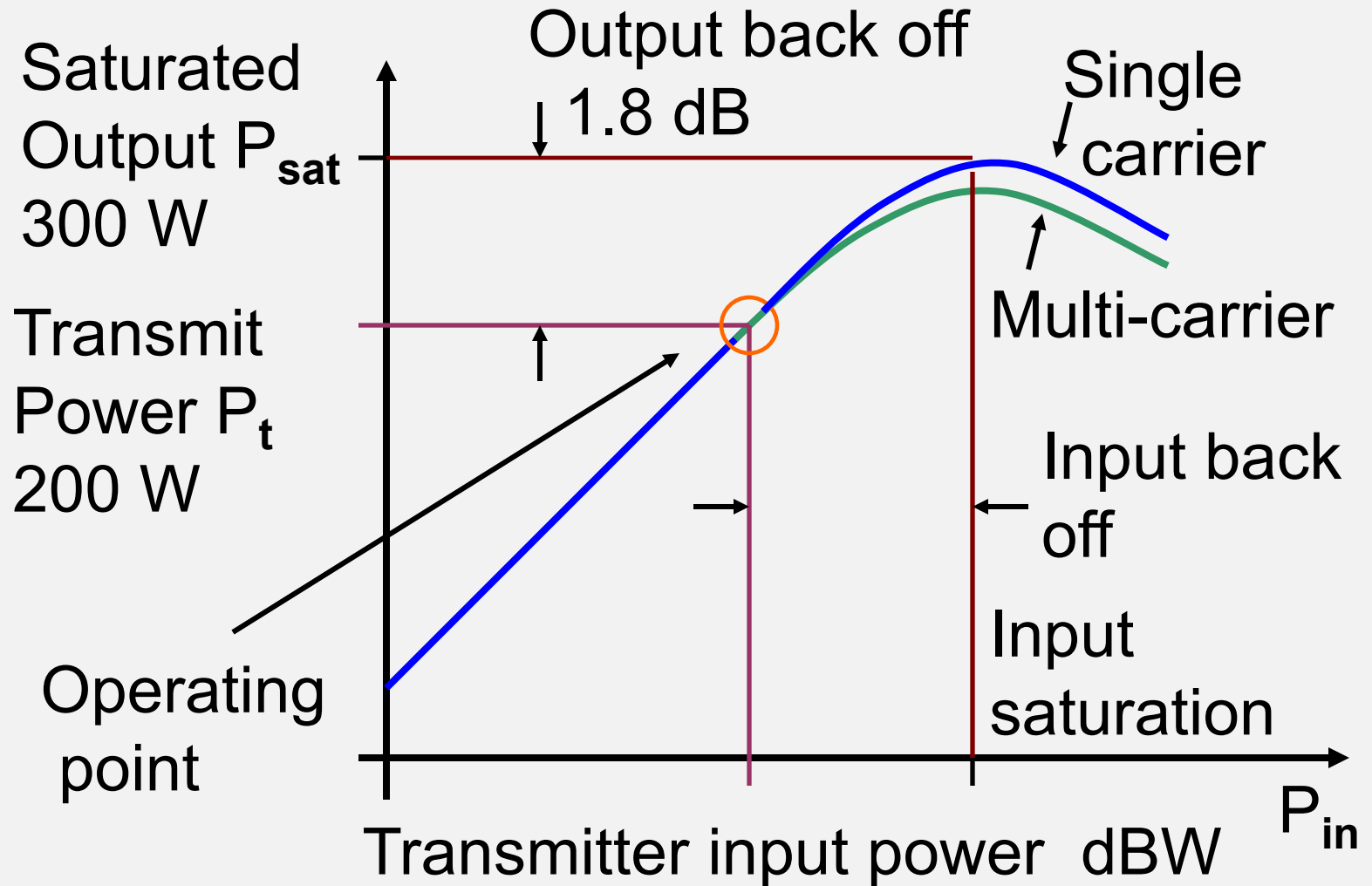
# C-Band Links

- The preceding design with 30 m Intelsat A earth stations gave very high C/N ratios
- Cost of earth stations is very high
- 30 m antenna has beamwidth  $\sim 0.1^\circ$  at 6 GHz
- Antenna must be steered to follow satellite motion of  $\pm 0.05^\circ$
- Costs can be lowered by using smaller antennas - Let's try this

# C-Band Links: Antenna Considerations

- Antenna cost increases as approximately antenna diameter raised to the power 2.7  
$$\text{Cost} \propto D^{2.7}$$
- A **10 m** antenna costs  $1/3^{2.7} \sim 1/20$  the cost of a 30 m antenna ( \$250 k versus \$5 M )
- Boost E/S uplink transmit power to compensate, from 100 W to  $P_r = 300$  W (sat)
- Apply back off: For a 300 W earth station, transmitter operation at **200 W** has 1.8 dB of back off

# Earth Station Transmitter Back-off



# C-band Links: Revised Design

- With the large antennas and high C/N ratios in transponder and receiving E/S we had:

$$D_t = D_r = 30 \text{ m}, G_t = 63 \text{ dB}, G_r = 60 \text{ dB}$$

$$P_t = 100 \text{ W} = 20.0 \text{ dBW}, \text{ EIRP} = 83.0 \text{ dBW}$$

- We could try 10 m E/S antennas with lower gain and boost transmit power

$$D_t = D_r = 10 \text{ m}, G_t = 53 \text{ dB}, G_r = 50 \text{ dB}$$

$$P_t = 200 \text{ W} = 23 \text{ dBW}, \text{ EIRP} = 76.0 \text{ dBW}$$

- At the satellite  $P_r$  will be down by 7.0 dB

# C-band Links

- Still wish to operate the satellite transponders with 15 dBW of output
- Need to revise  $(C/N)_{up}$  and  $(C/N)_{dn}$  and find the new  $(C/N)_o$
- We could rewrite the entire uplink and downlink power budgets to account for new antenna gains and transmitted power
- Observe that noise powers don't change
- Quicker method: look at changes in C/N ratios

# Modified C-band Uplink

- Uplink power budget (EIRP decreased by 7 dB):
  - EIRP **76.0 dBW**
  - $L_p$  at 6 GHz (typical) - 200.0 dB
  - $G_r$  at satellite (edge of beam) 23.0 dB
  - Miscellaneous losses - 2.0 dB
- $P_r$  (-96 dBW) **- 103.0 dBW**
- $N = -126.0$  dBW,  $(C/N)_{up} =$  **23.0 dB** (30 dB)

# Modified C-band Downlink

- Downlink power budget with reduced  $G_r$  :

$P_t$  saturated 17.0 dBW

$G_t$  satellite 26.0 dB

Output back off - 2.0 dB

Downlink path loss (typical) - 196.0 dB

Receive terminal gain 50.0 dB

Edge of beam loss - 3.0 dB

Miscellaneous losses - 2.0 dB

$P_r$  (-100 dBW) - 110.0 dBW

# Modified C-band Downlink

- Previous calculation for receiving earth station gave  $N = -134.7 \text{ dBW}$  (same)

- Hence

$$\text{new } (C/N)_{\text{up}} = 23.0 \text{ dB} \quad (30 \text{ dB})$$

$$\text{new } (C/N)_{\text{dn}} = 24.7 \text{ dB} \quad (34.7 \text{ dB})$$

- Using the reciprocal formula,

$$\begin{aligned} \text{new } (C/N)_o &= 1 / [ 1/295 + 1/200 ] = 119 \\ &= 20.8 \text{ dB} \quad (28.7 \text{ dB}) \end{aligned}$$

## Modified C-band Link

- Overall C/N is lowered by 7.9 dB with 10 m antennas to 20.8 dB
- We are still 2.8 dB over the minimum  $(C/N)_o$  of 18.0 dB, which was the spec!
- Quick method to find new overall  $(C/N)_o$ :
  - We reduced uplink EIRP from 83.0 to 76.0 dB
  - $(C/N)_{up}$  will be  $83.0 - 76.0 = 7.0$  dB lower
  - $(C/N)_{up} = 30.0 - 7.0 = 23.0$  dB

## Modified C-band Link

- We reduced receiving antenna gain by 10 dB:
  - $(C/N)_{dn}$  will be 10.0 dB lower
  - $(C/N)_{dn} = 34.7 - 10.0 = 24.7$  dB
- Quick method gives the same results for  $(C/N)$
- And  $(C/N)_o = 20.8$  dB as before
- We use decibels because of the ease with which a link design can be modified using the quick method

## Modified C-band uplink

- Check on transponder gain in redesign
- New power received (7 dB↓)  $P_r = -103.0$  dBW
- Transponder output power still  $P_t = +15$  dBW
- Higher transponder gain now required:  
$$G_{xp} = +15 - (-103) = 118 \text{ dB} \quad (111 \text{ dB})$$
- Gain must be increased by 7 dB
- Might need to review transponder design

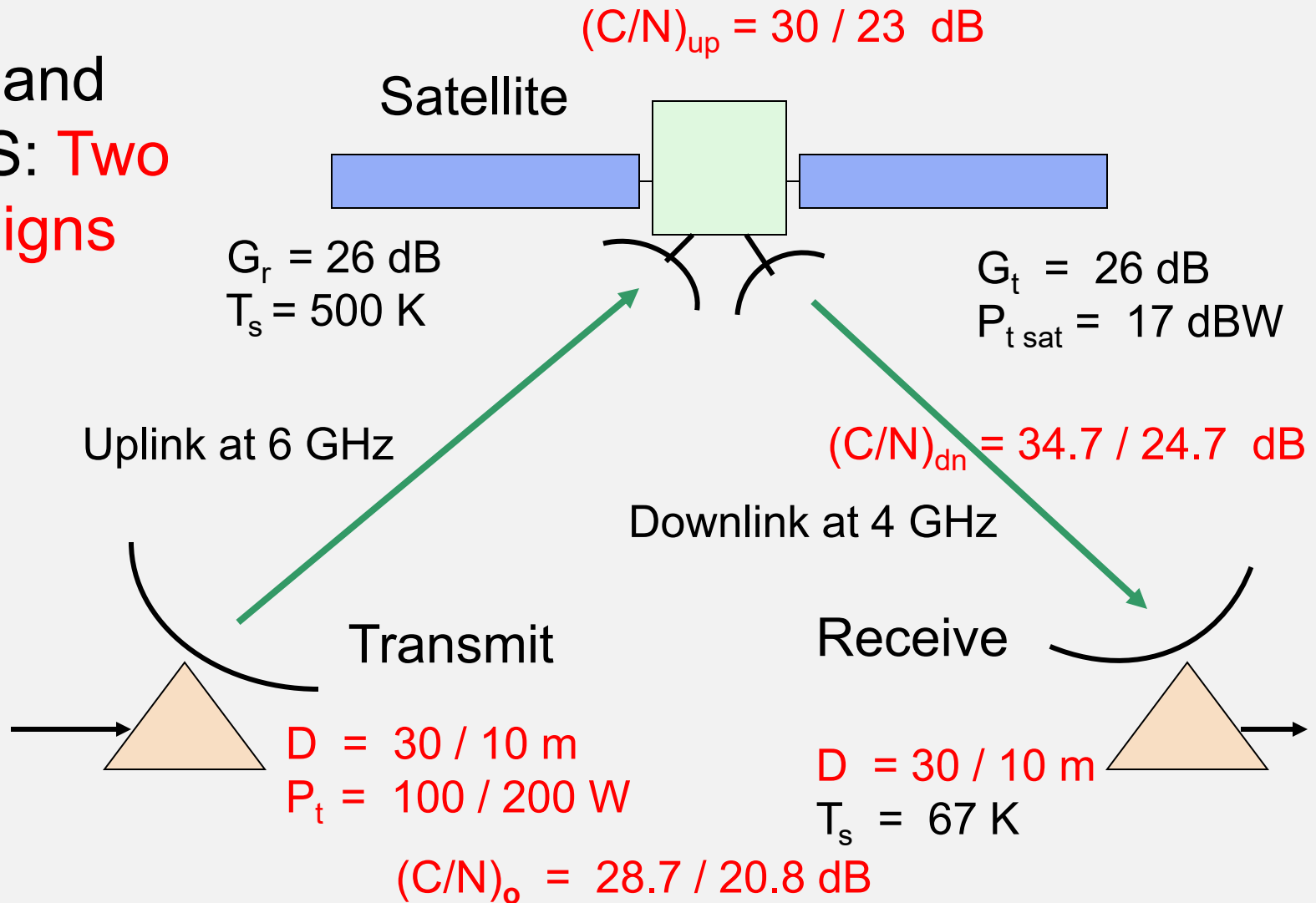
## Modified C-band uplink

- Check on antenna performance
- The diameter decreased by a factor of 3 which decreases the area by a factor of 9
- Cost is greatly reduced (x20!)
- Beamwidth increases to  $9 \times 0.13^\circ = 1.2^\circ$
- With more tolerance on direction, steering and tracking will be much easier
- But we did lose 7.9 dB in overall C/N

## Modified C-band uplink: Review

- We propose reducing the size of the E/S antennas from 30 m to 10 m, and raising transmit power by 3 dB
- Large savings in cost and complexity
- Our margin on  $(C/N)_o$  has dropped from 10 dB to 2.8 dB
- We still meet spec but are more vulnerable to variations in performance and weather
- Consult, consider trade-offs, iterate, and decide!

# C-band FSS: Two designs



Intelsat Atlantic Ocean Region satellite link  $(C/N)_o \geq 18.0 \text{ dB}$



# Rain Attenuation Margins

- The values of overall C/N for the two designs were obtained assuming clear air
- When rain affects the links, C/N values will be reduced
- Need to re-calculate uplink and downlink margins
- A *margin* is the amount by which a value can be reduced before hitting a limiting condition
- Limiting condition is still  $(C/N)_o = 18.0 \text{ dB}$

# Rain Attenuation Margins

- Question:

How much can  $(C/N)_{\text{up}}$  or  $(C/N)_{\text{dn}}$  be reduced before  $(C/N)_o = 18.0 \text{ dB}$  ?

- That is, how much rain can we tolerate on either link?
- We will discuss the amount of rain in terms of  $A \text{ dB}$ , the total attenuation on the signal due to rain on a path
- With a rain model we can relate  $A \text{ dB}$  to rain type, rate, and path length

# Rain Attenuation Margins

- We can assume that the occurrence of heavy rain on the uplink and downlink are independent events
- If the uplink is affected by heavy rain, assume downlink is in clear air and vice-versa
- (Not true for a *loop-back test*)

# Uplink Rain Attenuation Margin

- Uplink rain attenuation  $A_{up}$  dB:

- Power received at transponder input is

$$P_r \text{ rain} = P_r \text{ clear air} - A_{up} \text{ dBW}$$

$$(C/N)_{up} \text{ rain} = (C/N)_{up} - A_{up} \text{ dB}$$

- Assuming linear transponder operation

$$P_t \text{ rain} = P_t \text{ clear air} - A_{up} \text{ dBW}$$

$$(C/N)_{dn} \text{ rain} = (C/N)_{dn} - A_{up} \text{ dB}$$

# Uplink Rain Attenuation Margin

- Note that this result is true of a *linear* transponder for which  $P_t$  decreases by the same amount as  $P_r$
- For a *non-linear* transponder the decrease in  $P_t$  will generally be less
- For a *regenerative* transponder, or one that utilizes automatic gain control (AGC),  $P_t$  might not drop at all

# Uplink Rain Attenuation Margin

- For a linear transponder the C/N on both the uplink and downlink is reduced by  $A_{up}$
- This can be worked through the reciprocal formula to show that  $(C/N)_o$  is reduced by  $A_{up}$
- Conclusion:

With rain on the uplink and a linear transponder  $(C/N)_o$  is reduced by exactly  $A_{up}$  dB (the uplink rain attenuation)

# Uplink Rain Attenuation Margin

- The uplink rain attenuation margin is

$$\text{Uplink margin} = (C/N)_o \text{ clear air} - (C/N)_{o \text{ min}}$$

- In the C-band link example with the smaller 10 m antennas

$$(C/N)_o \text{ clear air} = 20.8 \text{ dB}, (C/N)_{o \text{ min}} = 18.0 \text{ dB}$$

- Uplink rain attenuation margin is 2.8 dB ( $\sim 3$  dB)
- Whether this is enough will depend on rainfall conditions

# Uplink Rain Attenuation Margin

- For the uplink, the margin on rain attenuation is the same as the margin on  $(C/N)_o$  – for a linear transponder
- If we can tolerate a drop of 3 dB in  $(C/N)_o$  then we can tolerate rain attenuation of 3 dB on the uplink
- This is not the case for the downlink – the margin on rain attenuation will be less than the margin on  $(C/N)_o$

# Downlink Rain Attenuation Margin

- For the downlink, there is (of course) direct attenuation of the power in the carrier wave
- But there is also a rise in noise power in the receiver because rain on the downlink increases the sky noise temperature
- These two factors generally lower the C/N on the downlink more than would be the case for the same rain attenuation on the uplink

# Downlink Rain Attenuation Margin

- The overall link is typically more vulnerable to rain on the downlink
- Next, we will apply our noise model for a lossy medium to determine the rise in noise power, the resulting value of  $(C/N)_{dn}$ , the impact on  $(C/N)_o$ , and the margin for rain attenuation on the downlink

